

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No.....

EC-703-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Optical Communication

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Give a brief introduction on OFC. Discuss about the optical spectral bands.
b) Write and explain the configuration of optical fiber modes.
2. a) What do you mean by a helical skew ray path within a graded index fiber?
b) An optical fiber in air has an NA of 0.4. Compare the acceptance angle for meridional rays with that for skew rays which change direction by 100° at each reflection.
3. a) Explain the source to fiber power launching. Write some practical applications of this type of launching.
b) Explain the process of fiber splicing. What precautions should be kept in mind during fiber splicing?
4. a) Determine the expression of photo detector noise.
b) Discuss different types of losses and distortion in optical fibers.

EC-703-GS

PTO

[2]

5. Define the quantum efficiency and the responsivity of a photo detector. Derive an expression for the responsivity of an intrinsic photodetector in terms of quantum efficiency of the device and the wavelength of the incident radiation.
6.
 - a) Explain the three main factors limit the speed of response of a photodiode.
 - b) What do you know of p-n and p-i-n photodiode receiver?
7. With the help of block diagram explain the working of fundamental receiver operation also show the shape of a digital signal at different points along an optical link.
8.
 - a) Explain in detail equilibrium mode simulation by mode filtering:
 - i) Mandrel wrap
 - ii) Dummy fiber
 - b) What are the basic applications and types of optical amplifiers discuss?
